Ibadan Urban Heat Risk Intelligence System

Summary Report | 2023

Urban Heat Island intensity mapping, spatial autocorrelation, GWR heat-driver modelling, and thermal vulnerability assessment for Ibadan Metropolis, Oyo State, Nigeria.

Methodology Conceptualisation

- Landsat dry-season LST is used as the thermal exposure baseline.
- NDVI, NDBI, and GHSL built-up density are used as heat-driver indicators.
- Moran's I, LISA, and Getis-Ord Gi* identify spatial clustering of heat.
- GWR uses a 500 m grid and forced adaptive bandwidth for stable local modelling.
- HVI combines exposure, sensitivity, and adaptive capacity indicators.
- OSM green/water distance layers are treated as static adaptive-capacity proxies.

Key Raster Statistics

Land Surface Temperature: min=28.065, mean=34.932, max=47.793, std=3.362

Urban Heat Island Intensity: min=-4.845, mean=2.022, max=14.884, std=3.362

Normalized Difference Vegetation Index: min=-0.194, mean=0.427, max=0.814, std=0.131

Normalized Difference Built-up Index: min=-0.414, mean=-0.063, max=0.365, std=0.114

Heat Vulnerability Index: min=0.047, mean=0.294, max=0.771, std=0.120

Heat Exposure Index: min=0.000, mean=0.348, max=1.000, std=0.170

Sensitivity Index: min=0.000, mean=0.048, max=0.881, std=0.104

Adaptive Capacity Index: min=0.131, mean=0.450, max=0.890, std=0.128

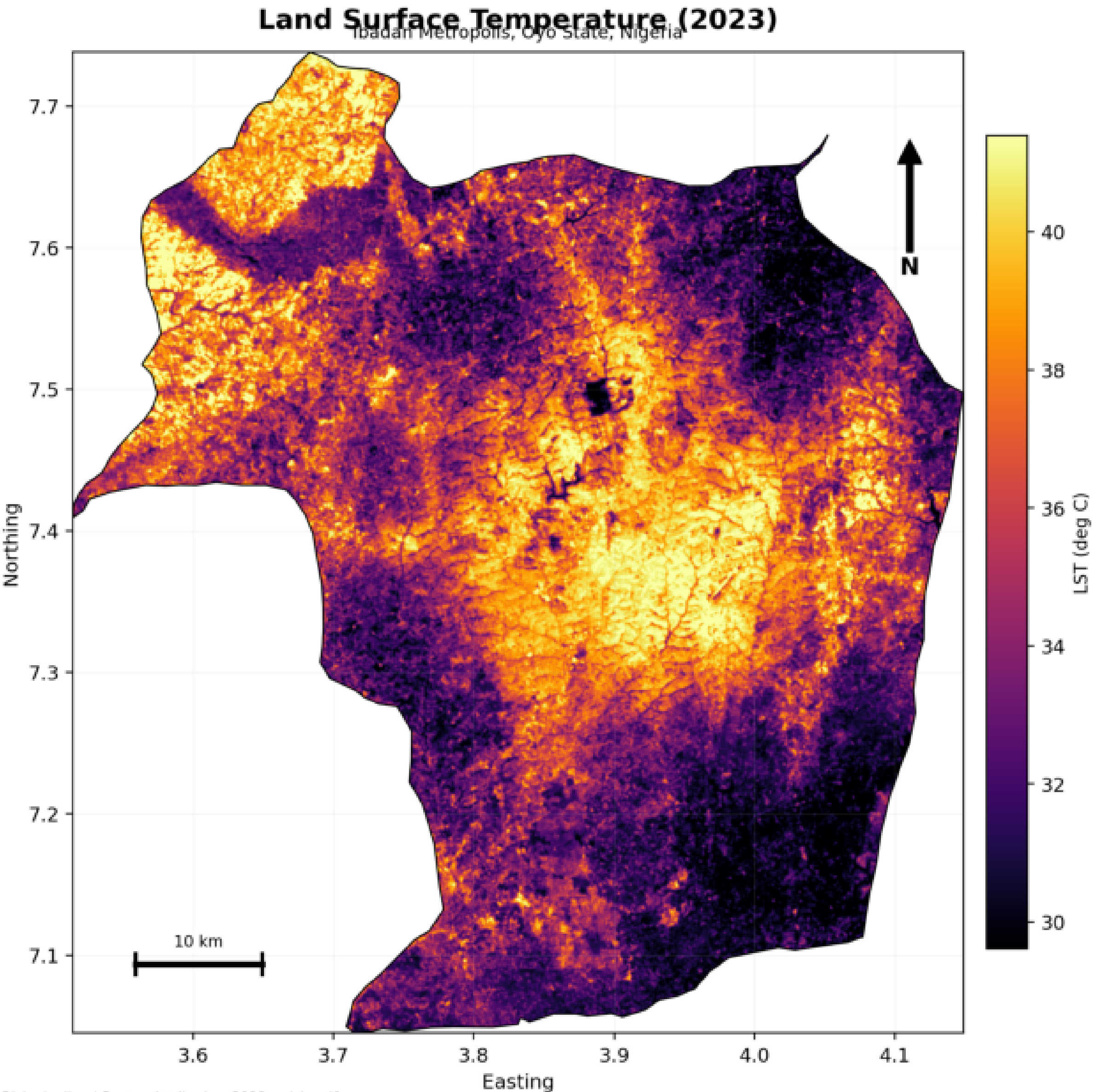
Result Interpretation

- Higher LST, UHI, HVI, and hot-spot membership indicate priority heat-risk zones.
- Negative NDVI coefficients imply vegetation cooling effects.
- Positive NDBI and built-up density coefficients imply urban surface warming effects.
- LGA rankings support planning prioritisation but should be validated locally.

Referenced Output Tables

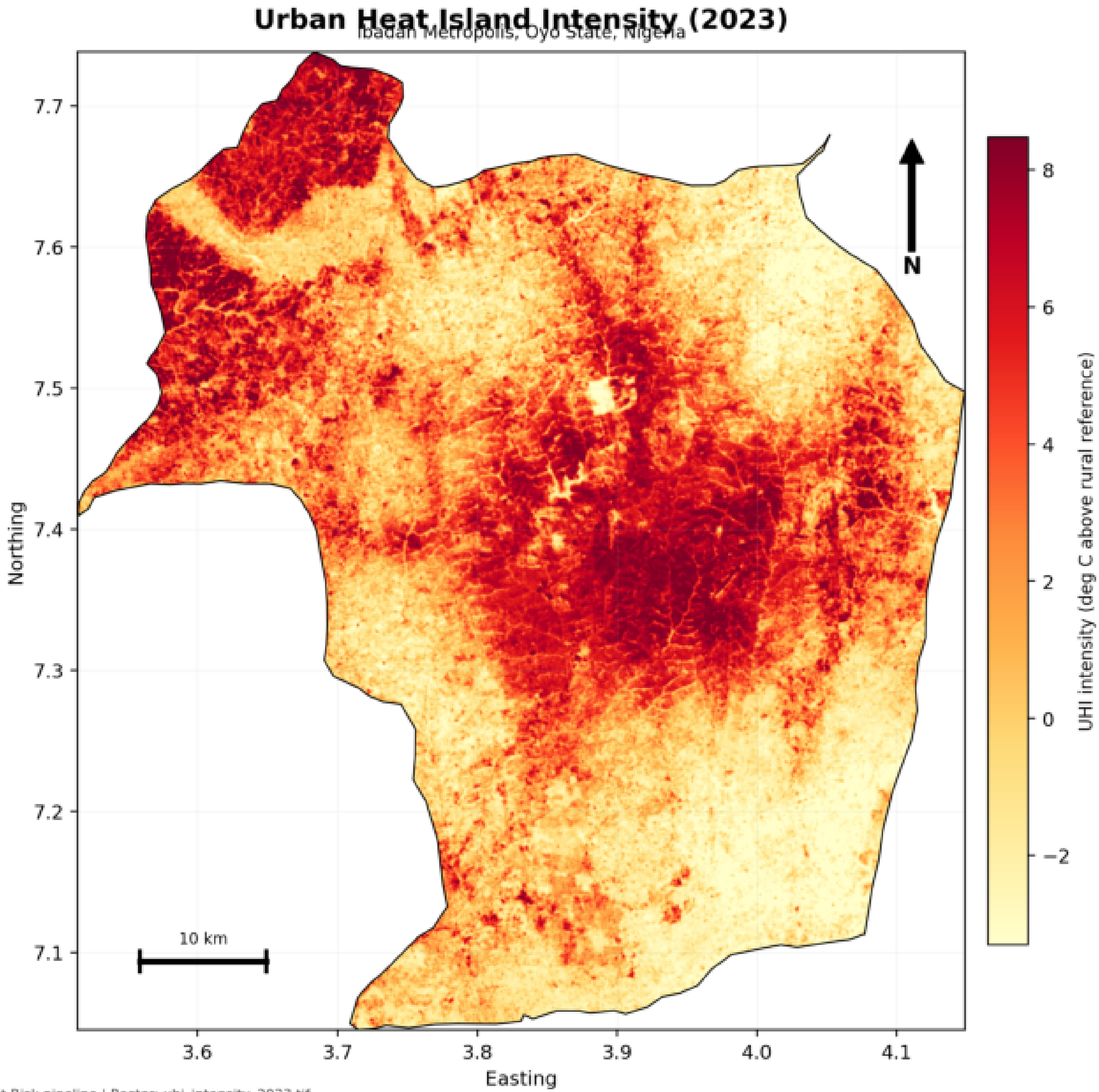
- lga_lst_summary: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/lga_lst_summary_2023.csv
- lga_hvi_summary: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/lga_hvi_summary_2023.csv
- hvi_ranking: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/ibadan_hvi_ranking_2023.csv
- morans_i: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/morans_i_2023.csv
- gwr_correlations: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/gwr_input_correlations_2023.csv
- gwr_vif: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/gwr_input_vif_2023.csv
- gwr_ols_coefficients: C:/Users/Marvii/Documents/GIS_PROJECTS/ibadan-urban-heat-risk/outputs/tables/gwr_global_ols_coefficients_2023.csv

Lst



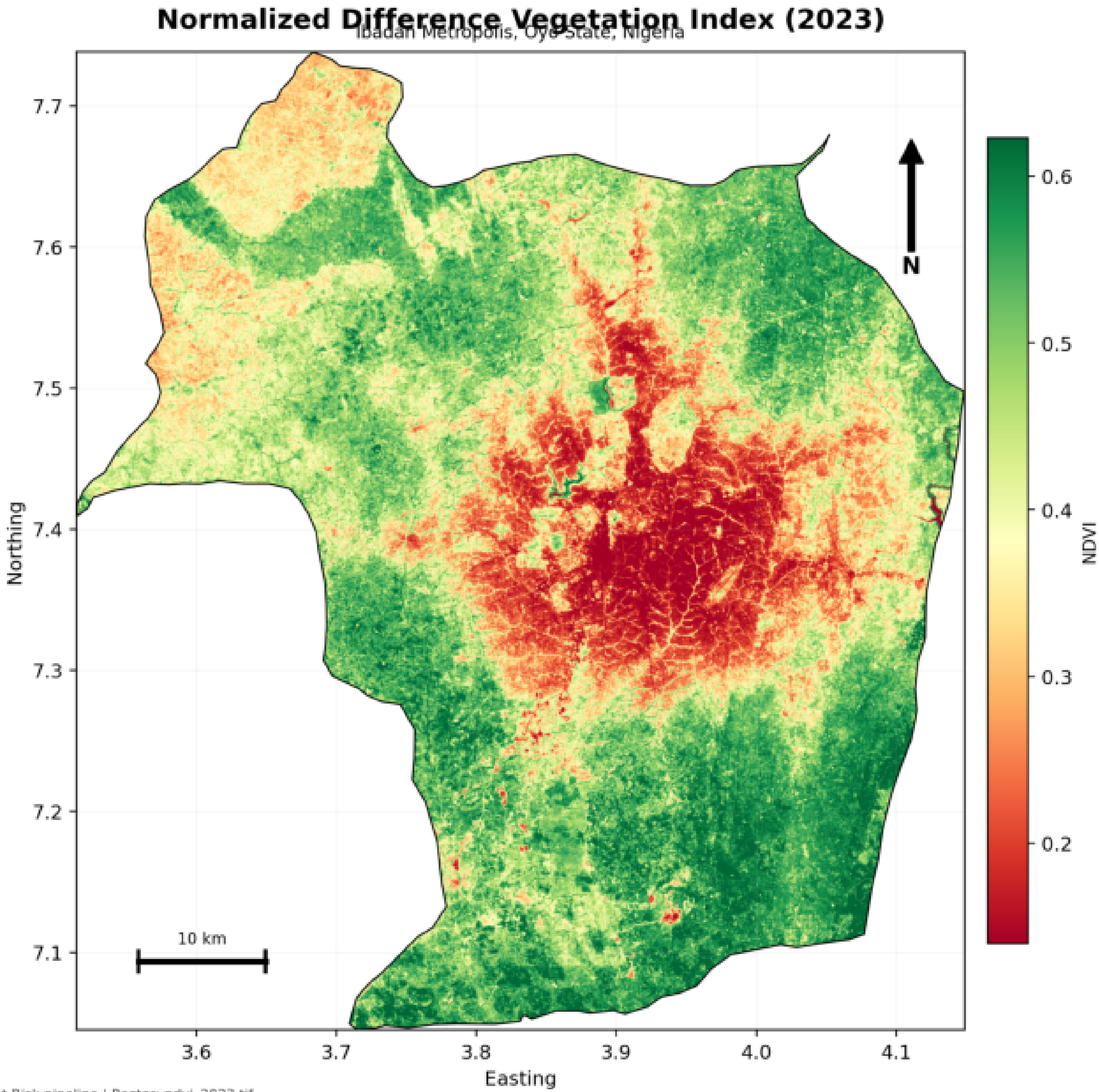
Source: Ibadan Heat Risk pipeline | Raster: lst_ibadan_2023_celsius.tif

Uhi



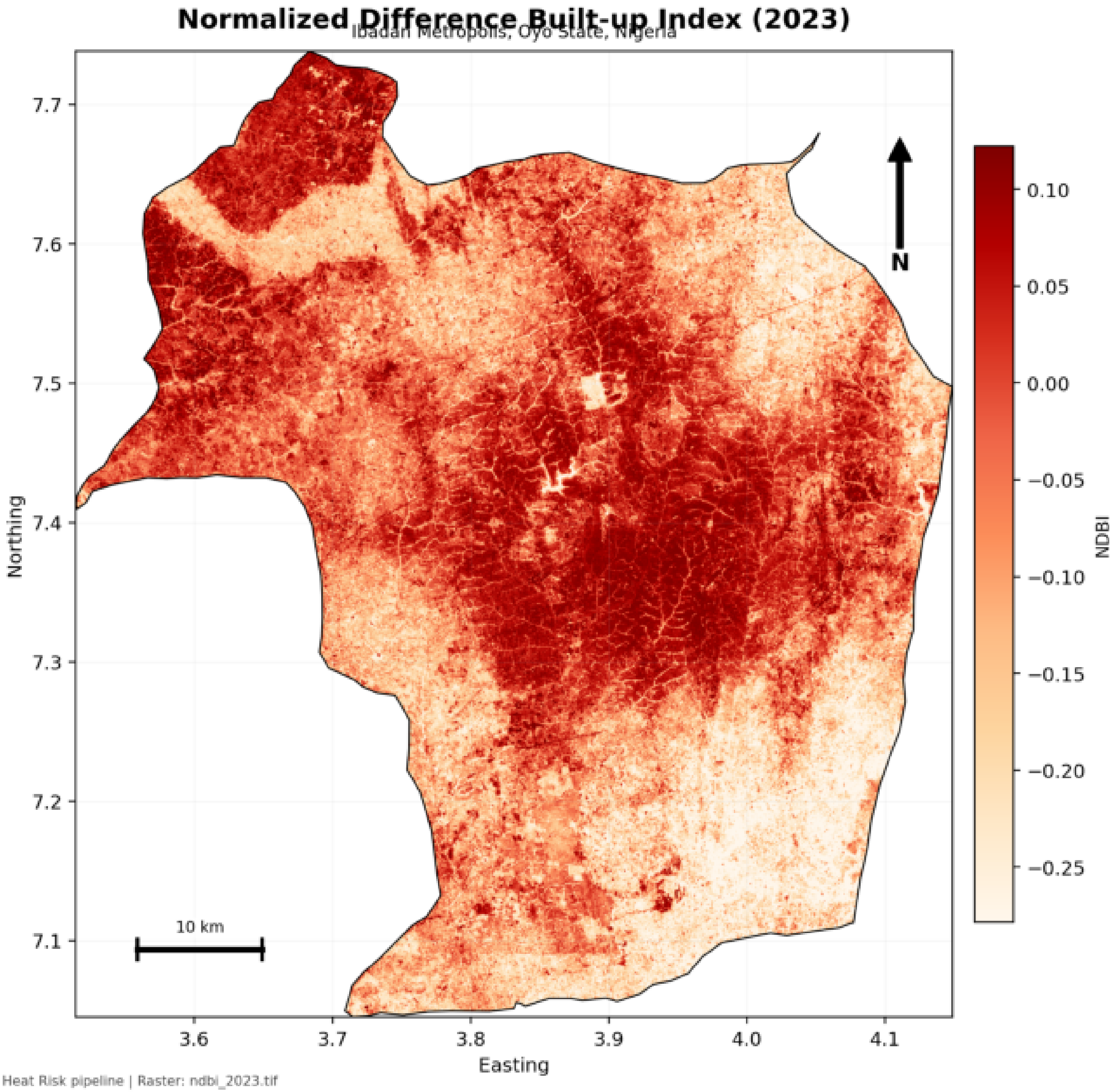
Source: Ibadan Heat Risk pipeline | Raster: uhi_intensity_2023.tif

Ndvi



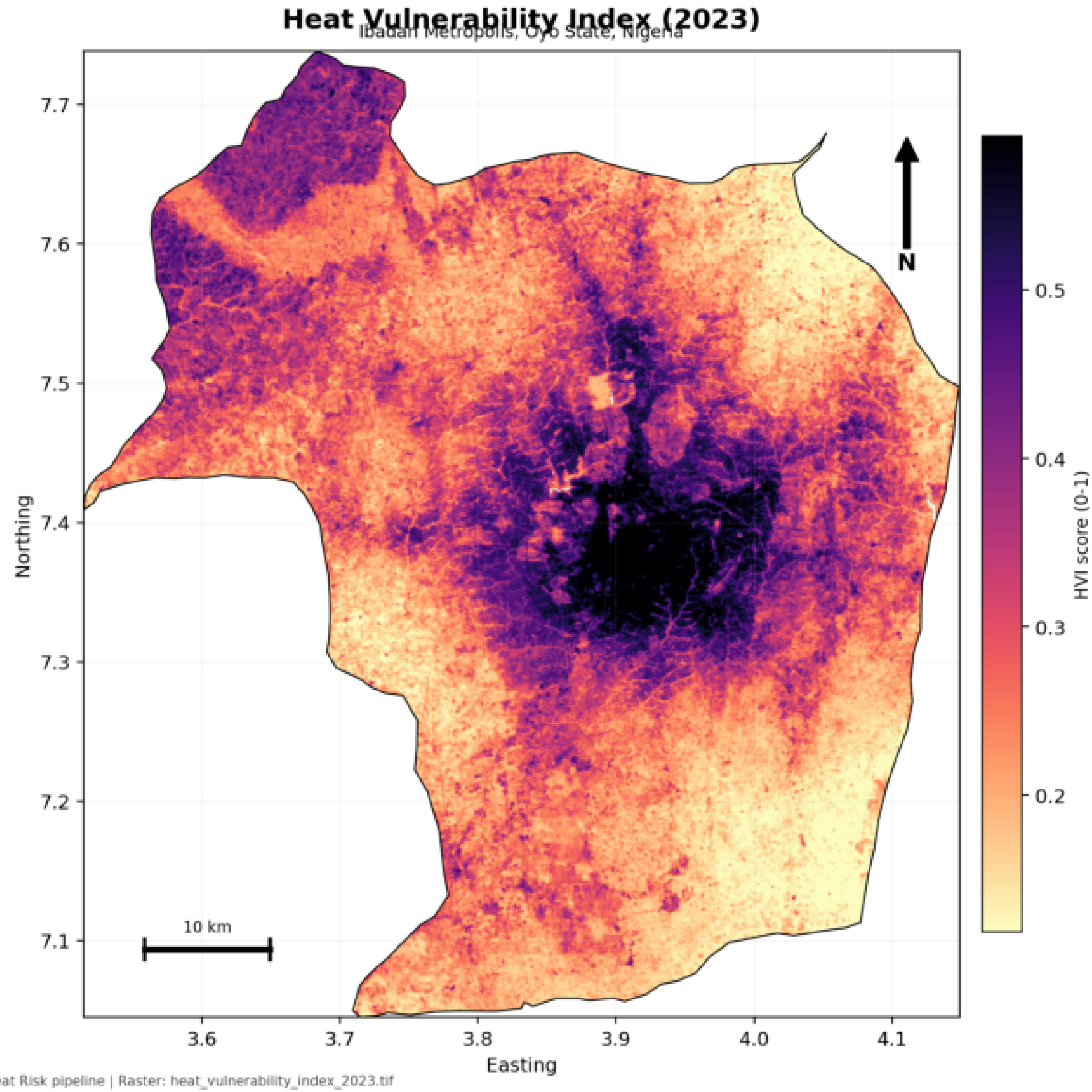
Source: Ibadan Heat Risk pipeline | Raster: ndvi_2023.tif

Ndbi



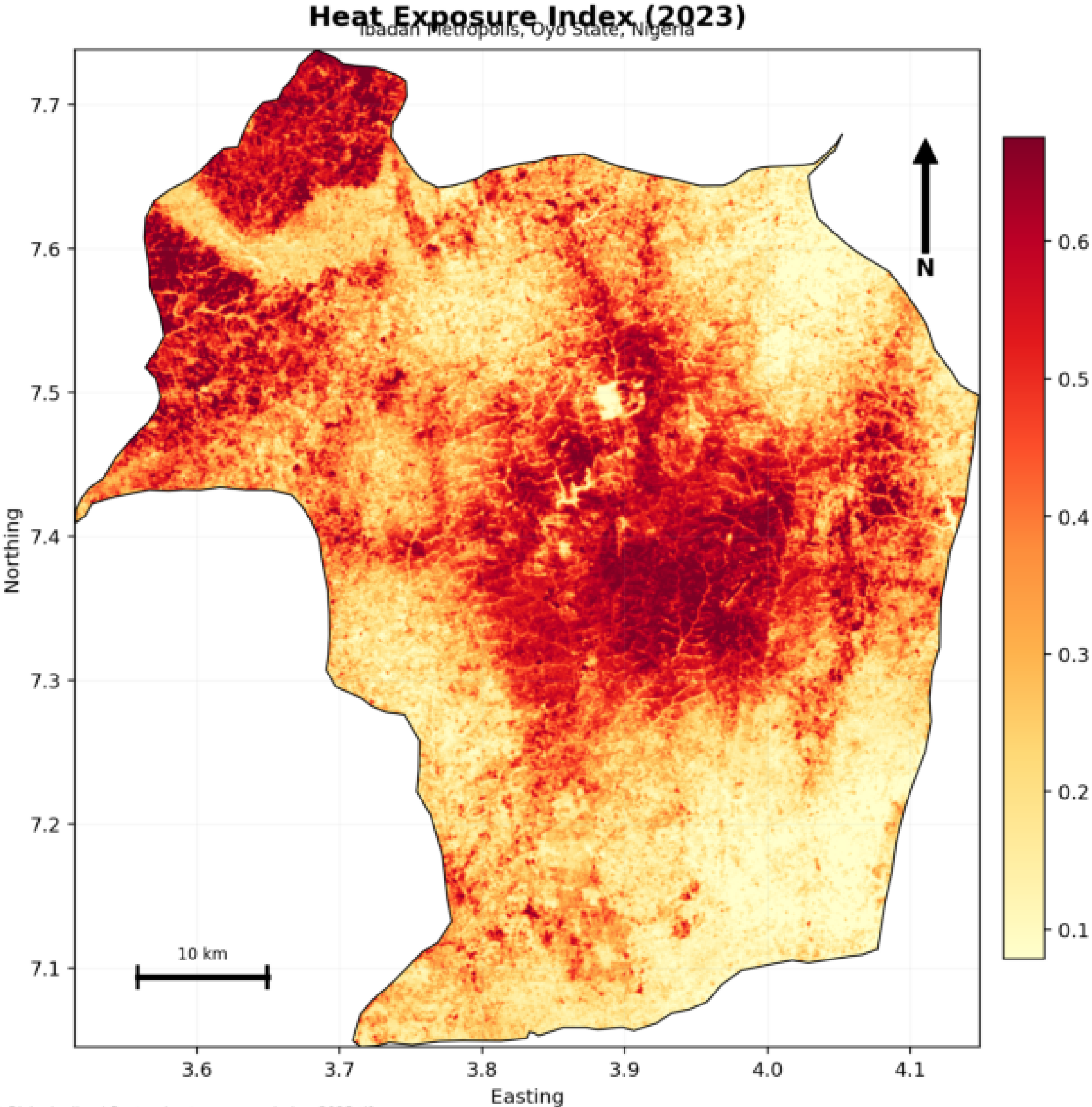
Source: Ibadan Heat Risk pipeline | Raster: ndbi_2023.tif

Hvi



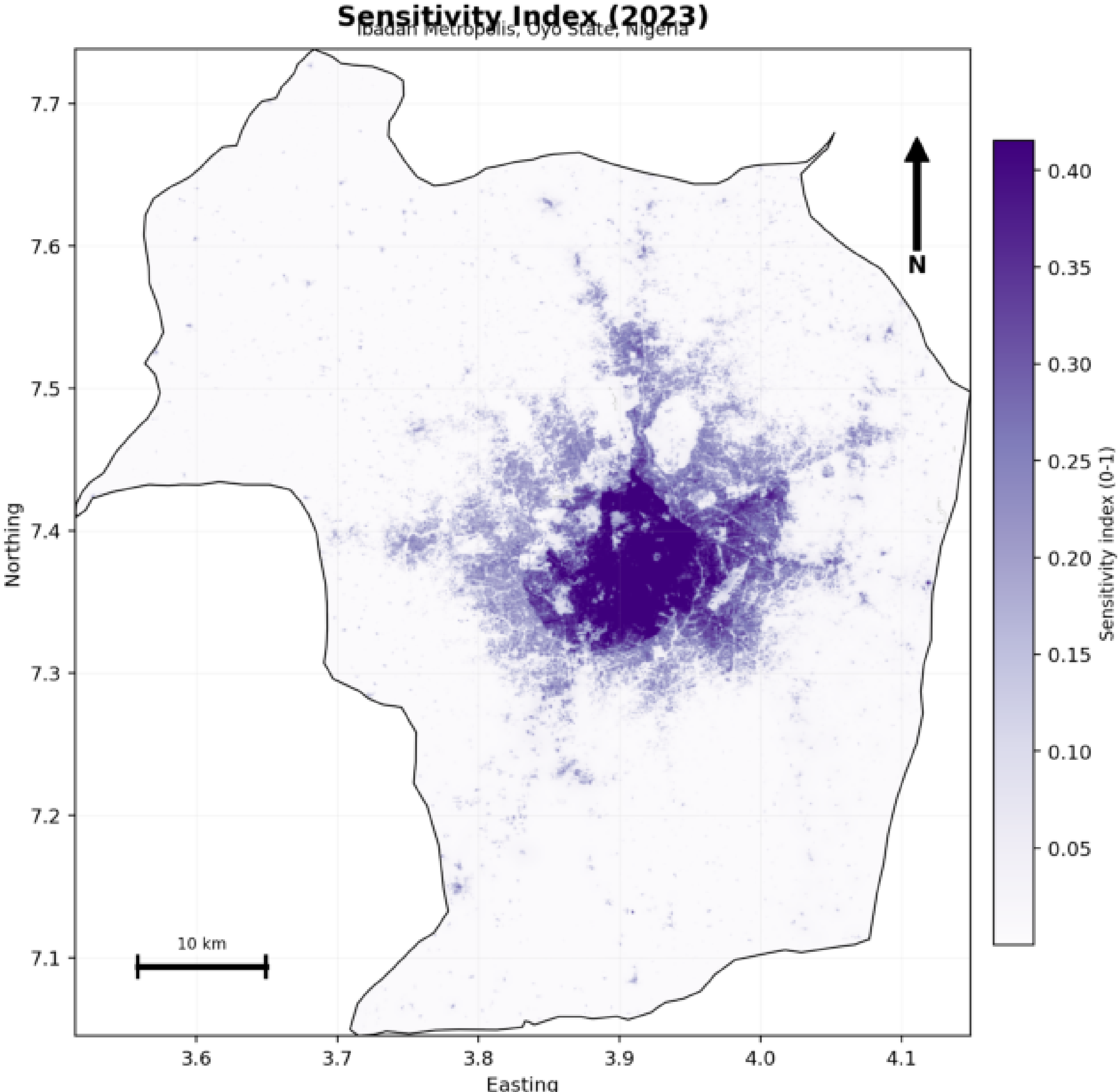
Source: Ibadan Heat Risk pipeline | Raster: heat_vulnerability_index_2023.tif

Heat Exposure



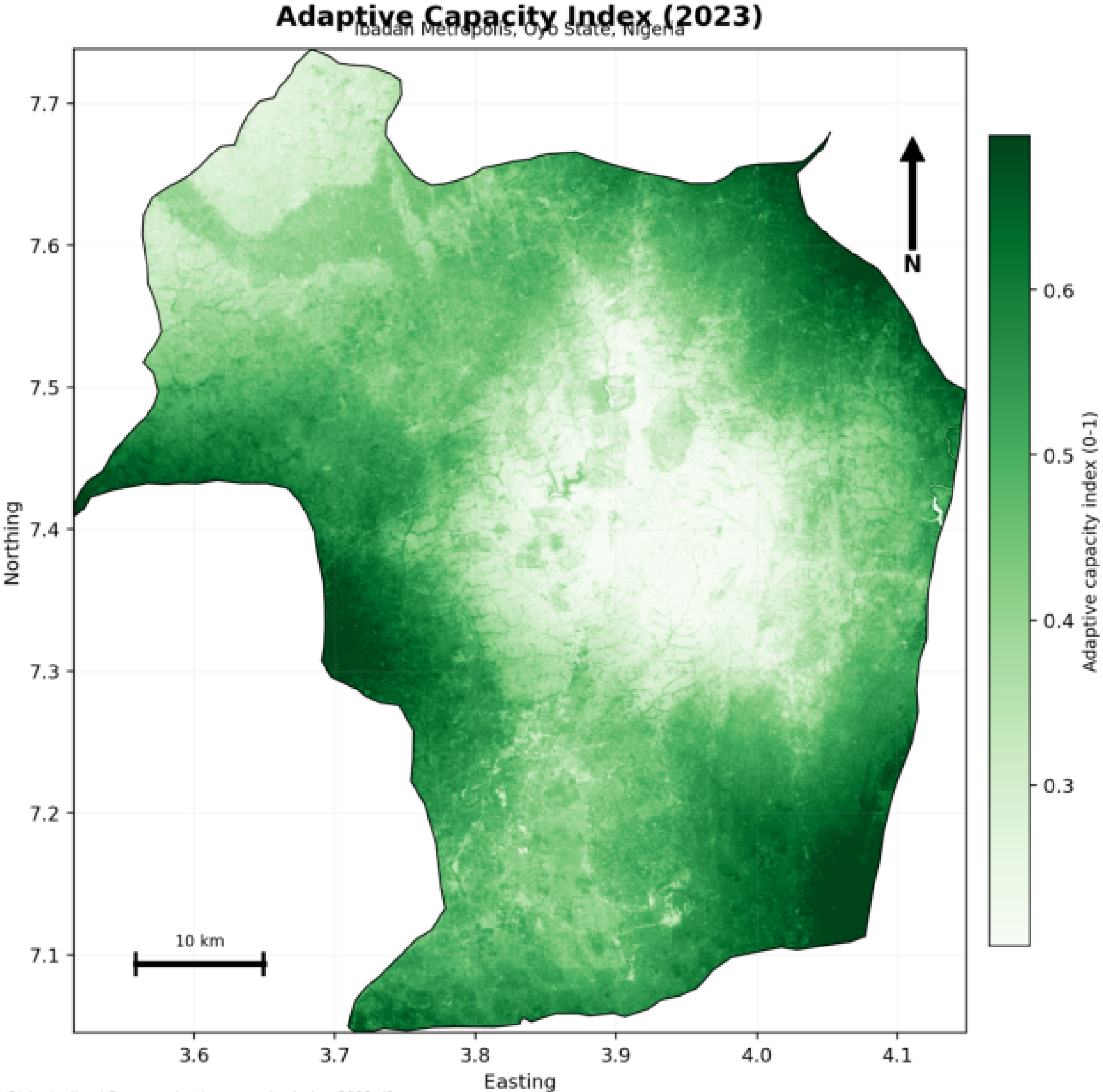
Source: Ibadan Heat Risk pipeline | Raster: heat_exposure_index_2023.tif

Sensitivity



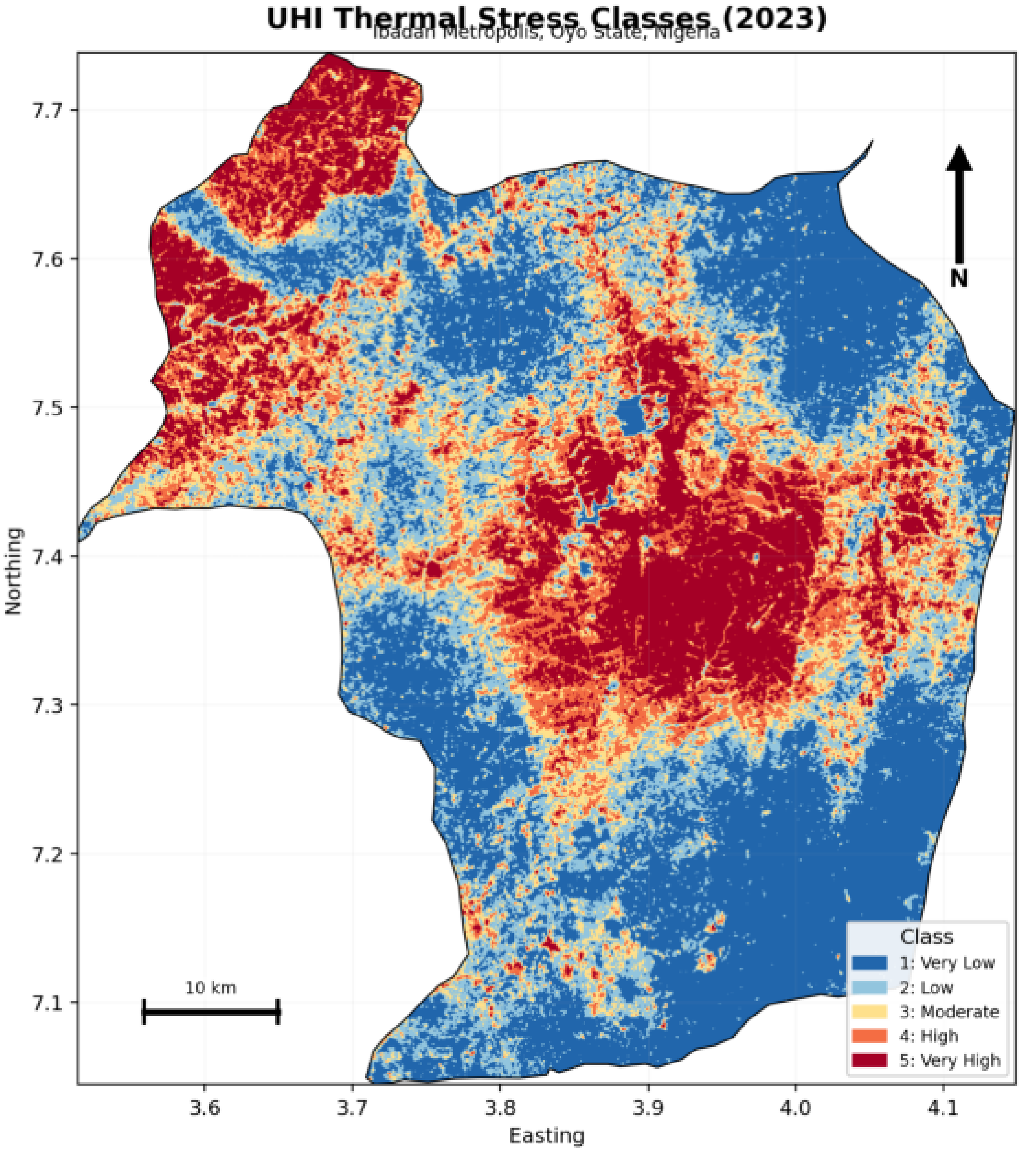
Source: Ibadan Heat Risk pipeline | Raster: sensitivity_index_2023.tif

Adaptive Capacity



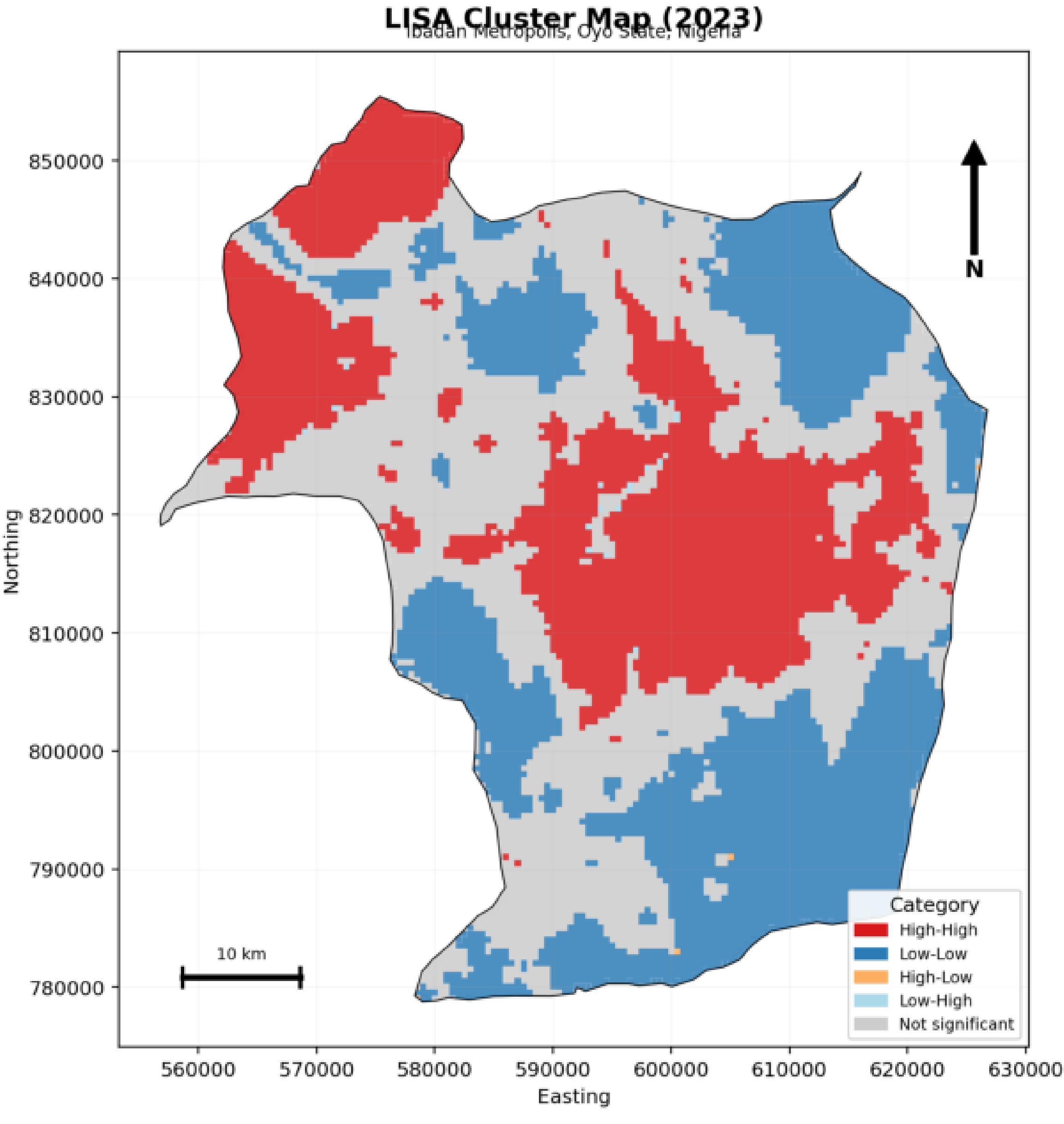
Source: Ibadan Heat Risk pipeline | Raster: adaptive_capacity_index_2023.tif

Uhi Classes

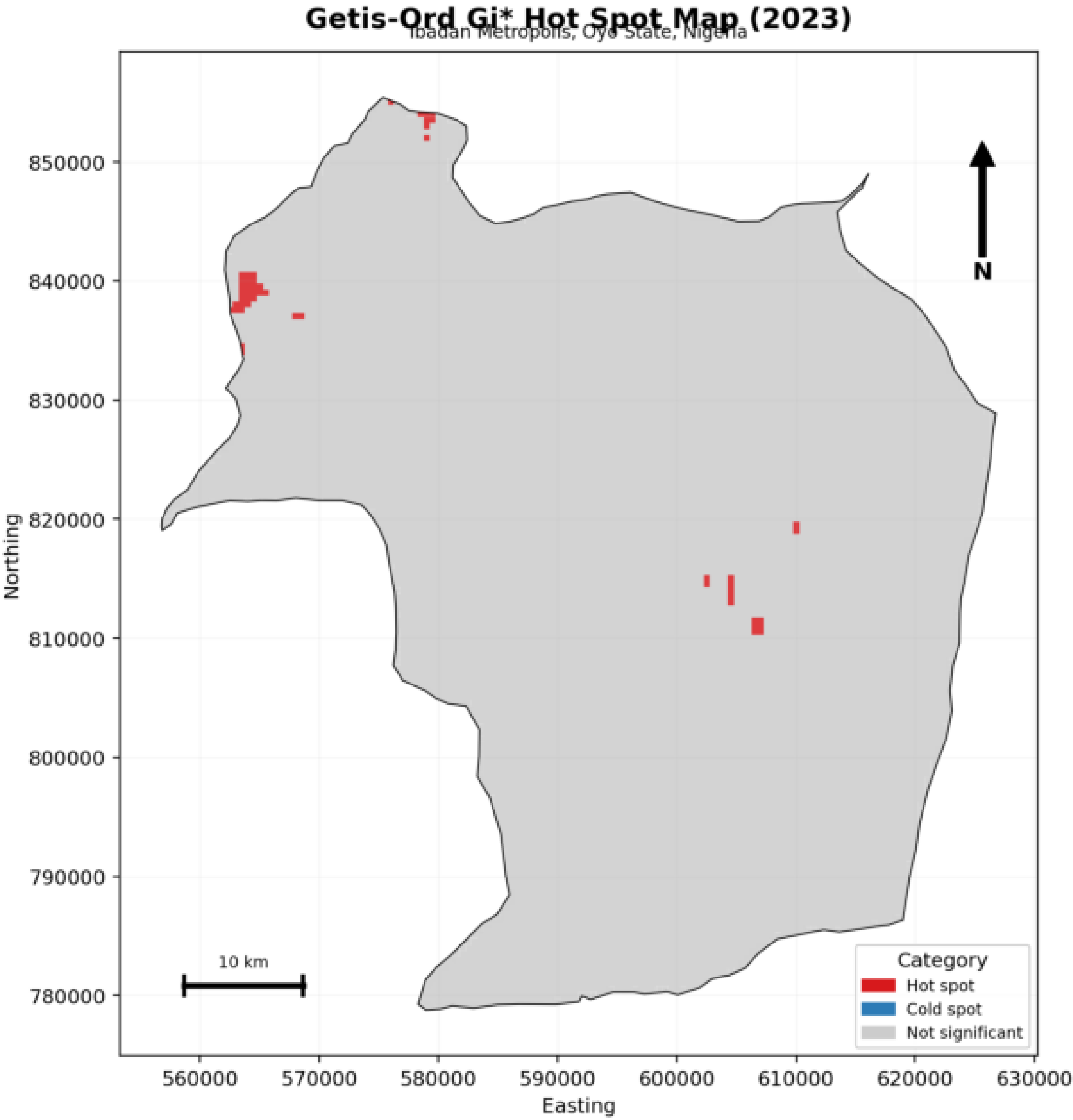


UHI Thermal Stress Classes (2023)
Ibadan Metropolis, Oyo State, Nigeria
Northing
7.7
7.6
7.5
7.4
7.3
7.2
7.1
Easting
3.6
3.7
3.8
3.9
4.0
4.1
N
10 km
Class
1: Very Low
2: Low
3: Moderate
4: High
5: Very High

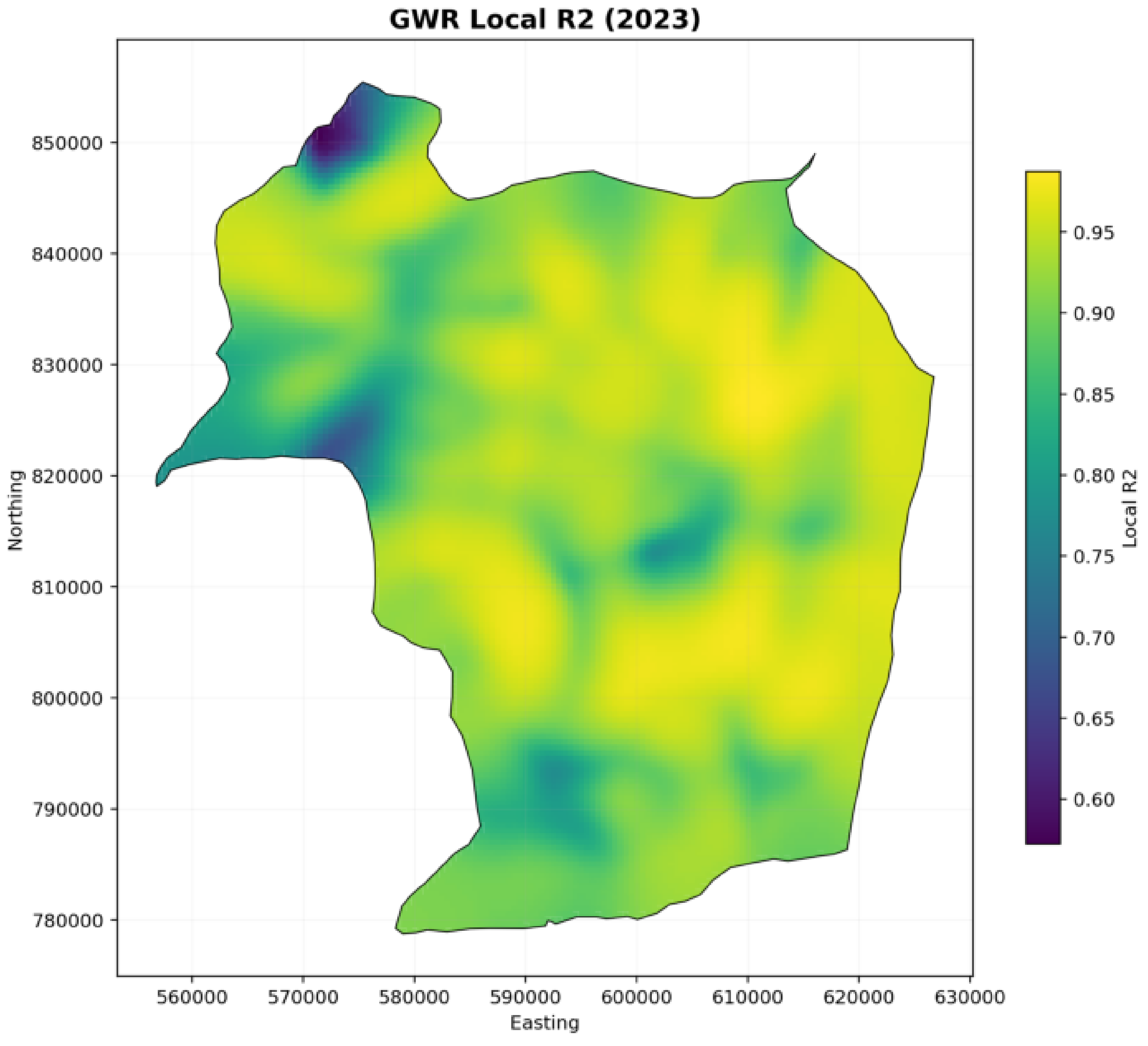
Lisa Clusters



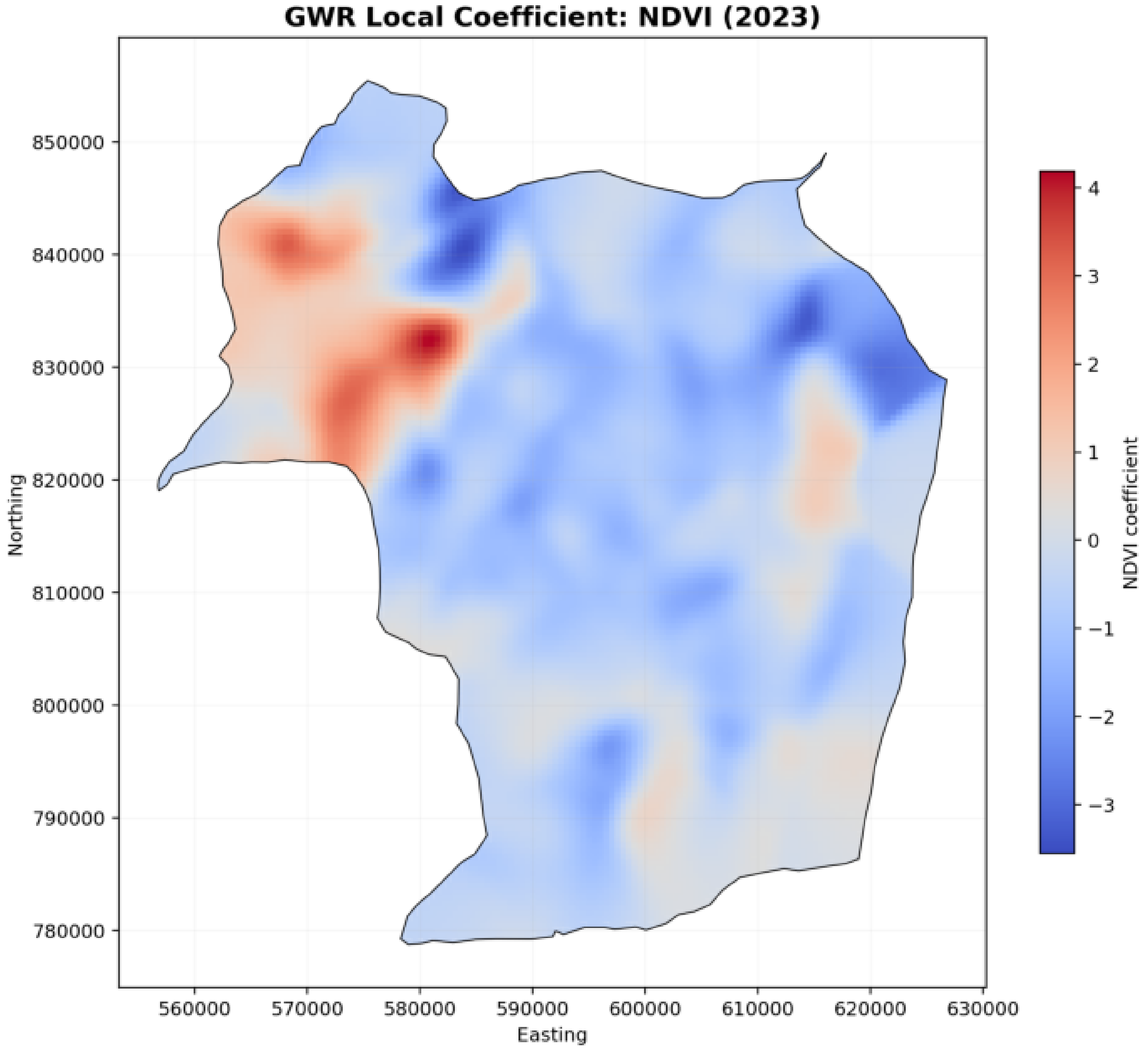
Gistar Hotspots



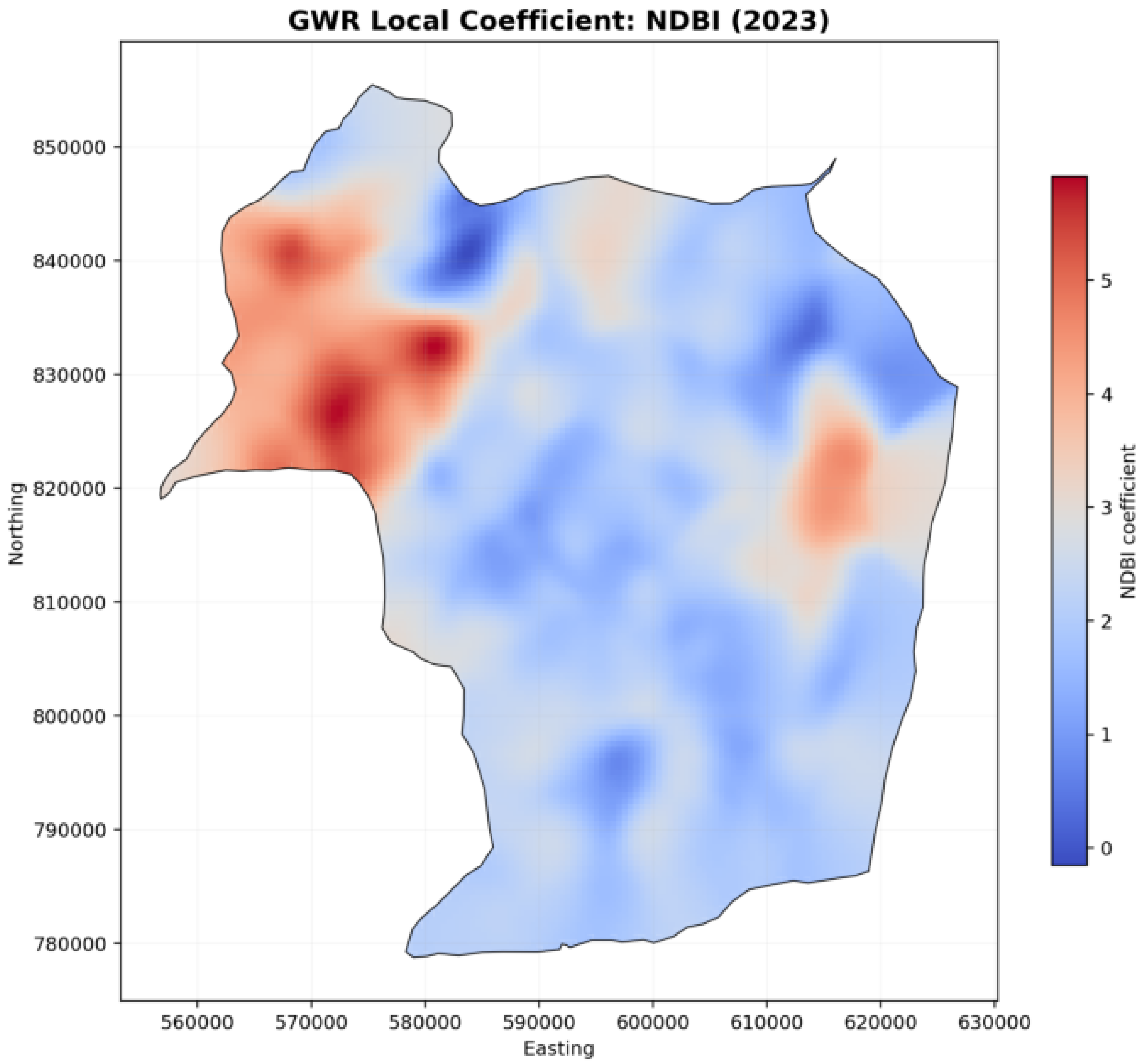
Gwr Local R2


GWR Local R2 (2023)
850000
840000
830000
820000
810000
800000
790000
780000
Northing
560000
570000
580000
590000
600000
610000
620000
630000
Easting
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
Local R2

Gwr Coef Ndvi



Gwr Coef Ndbi



Gwr Coef Built Up Density